

Machine Learning

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Supervised Learning

} Given dataset will have both
features and label
independent variables dependent variables

Regression

features: any kind of data

label: Continuous ND

Classification

features: any kind of data

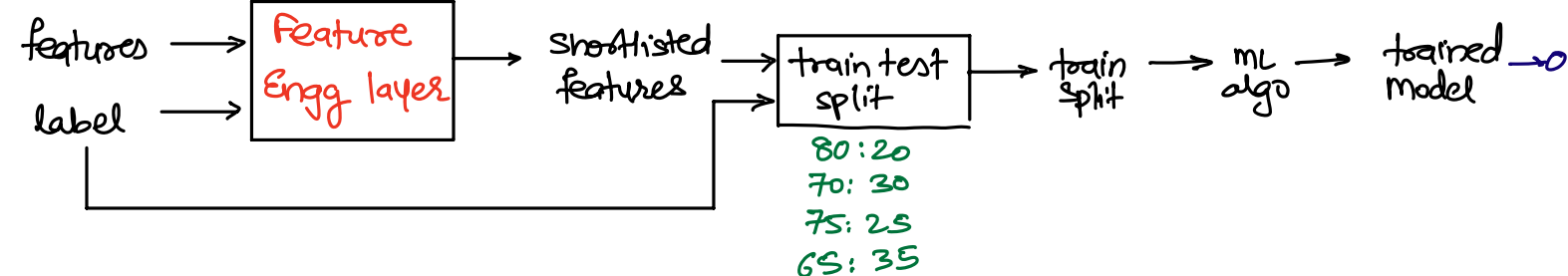
label: Discrete ND
-or-
Categorical Data

Model Building

Ensure the given dataset is compatible for Inferential Stats.

ML algo

model convergence
data convergence



test split

trained model

score

- check for Generalization
- check for CL criteria

Classification

Binary Classification

Given label column will have two unique values

Social Network Ads

Multi-class classification

Given label column will have more than two unique values

Iris example

Typical Classification Dataset

Balanced Dataset

Here the support of each unique label is equal.

-or-

support is even

-or-

support is equal

no. of records for each unique label is equal.

eg: iris dataset

setosa 50

versicolor 50

virginica 50

Unbalanced dataset

-or-

Imbalanced dataset

no of records for each unique label is different

eg. social network ads

0 → 257

1 → 143

Evaluation metrics for supervised learning

} ... check the quality of the converged/ model trained

Regression

• r^2 score
(Adjusted R^2)

Classification

- Accuracy
- Precision
- Recall
- F1-score

When to use which
metric for evaluating
a classification model?

if balanced Dataset

ACCURACY

- for generalization check
- for CL check with entire dataset^u

if imbalanced dataset

Guidelines by PN

Statistical
approach

(both binary classification
and multi-class classification)

Accuracy → to filter generalized
model

F1 Score → for CL check with
entire dataset.

$f1_score(\text{features}, \text{label}) \geq CL$

Domain-based
approach

(usually done for
Binary classification)

Accuracy → to filter generalized
model.

PR pair → for CL check with
entire dataset.
(Precision
Recall
pair)

Confusion Matrix (CM)

Error matrix in tabular form to **VISUALIZE** the performance of the classification model.

eg: emailSpam.csv.
100 records
ham
spam

			predicted values			
			ham	spam		
Actual labels	Ground Truth	Actual Values	ham	40	10	$40 + 10 = 50$ $\text{support}(\text{ham}) = \underline{50}$
Actual labels	Ground Truth	Actual Values	spam	5	45	$\text{support}(\text{spam}) = 50$

misclassified labels

no. of records model predicted wrong.

eg: iris.csv
150 records
Versicolor (Ve)
Virginica (Vi)
Setosa (Se)

		predicted values ^{m3}			
		Se	Ve	Vi	
ground truth	Se	41	4	5	$s(\text{Se}) = 50$
	Ve	0	50	0	$s(\text{Ve}) = 50$
	Vi	0	1	49	$s(\text{Vi}) = 50$

$\sum s_i = 150$

[Setosa, Versicolor, Virginica]

$s(\text{Ve}) = 50$
 $s(\text{Se}) = 50$
 $s(\text{Vi}) = 50$

} balanced dataset

$$41 + 50 + 49 = 140$$

misclassified records $\rightarrow 4 + 5 + 0 + 0 + 0 + 1$

How many number of records were there in the given dataset? $\rightarrow$

$$s(0) + s(1)$$

$$\sum s_i = 257 + 143$$

$$= 400$$

ground truth

	0	1
0	237	20
1	41	102

$$s(0) = 237 + 20$$

$$= 257$$

$$s(1) = 41 + 102$$

$$= 143$$

applicable for entire dataset

eg: Gmail Classification for Inbox data

Primary
Social
Promotions
Updates

predicted labels

Primary Promotions Social Updates

actual labels
ground truth

Primary
Promotions
Social
Updates

	Primary	Promotions	Social	Updates
Primary	40	1	0	1
Promotions	35	45	0	2
Social	0	0	10	3
Updates	2	1	4	20

$$s(Pr) = 42$$

$$s(Pro) = 82$$

$$s(Social) = 13$$

$$s(Updates) = 27$$

Correctly Predicted $\rightarrow 115$

Misclassified Records $\rightarrow 49$

• entire sample set.

• model is a generalized model.

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" given dataset is unbalanced

" if stat approach:

f1Score $\geq$ CL : approve

else

discard

if domain Approach

$PR \geq CL$: approx

else

discard.

Evaluation metrics for
classification



41	4	5
0	50	0
0	1	49

Accuracy: applicable for the entire model / dataset / all labels.
(go diagonal)

$$\begin{aligned} acc_{(model)} &= \frac{\sum \text{diagonal values}}{\sum \text{all values in CM.}} \\ &= \frac{41 + 50 + 49}{41 + 4 + 5 + 0 + 50 + 0 + 0 + 1 + 49} \\ &= \frac{140}{150} = 0.9333.. \end{aligned}$$

	Se	Ve	Vi
Se	41	4	5
Ve	0	50	0
Vi	0	1	49

Precision :- applicable for each unique label.
(go vertical)

$$P_{\text{setosa}} = \frac{\text{trueValue}}{\sum \text{total Selected Values}} = \frac{41}{41+0+0} = 1 //$$

$$P_{\text{versicolor}} = \frac{\text{trueValue}}{\sum \text{total Selected Value}} = \frac{50}{4+50+1} = 0.9 //$$

$$P_{\text{virginica}} = \frac{\text{trueValue}}{\sum \text{total Selected Value}} = \frac{49}{5+0+49} = 0.9 //$$

	Se	Ve	Vi
Se	41	4	5
Ve	0	50	0
Vi	0	1	49

Recall → applicable for each unique label.
(go horizontal)

$$R_{\text{(setosa)}} = \frac{\text{trueValue}}{\sum \text{total Selected Value}} = \frac{41}{41+4+5} = 0.82 //$$

$$R_{\text{(versicolor)}} = \frac{\text{trueValue}}{\sum \text{total Selected Value}} = \frac{50}{0+50+0} = 1 //$$

$$R_{\text{(virginica)}} = \frac{\text{trueValue}}{\sum \text{total Selected value}} = \frac{49}{0+1+49} = 0.98 //$$

Concept of PR pair } → Note: This is an exclusive **GUIDELINE** prepared by PN. Use it at your **RISK** :)

eg: Email Spam classification

ham
spam

	h	s
h	50	20
s	40	40

given cm of the generalized model. Goal is to check whether to approve the model for deployment or improve the model.

use Domain Approach.

$$s(\text{ham}) = 50 + 20 = 70$$

$$s(\text{spam}) = 40 + 40 = \underline{\underline{80}}$$

which of the following **MISTAKE** committed by the model is **NOT TOLERABLE** for the business?

	h	s
h	50 ₁	20 ₂
s	40 ₃	40 ₄

-
- ① ham ^{actual} predicted as ^{predicted} spam
 - ② spam predicted as ham

identified
PR pairs

$$P_{\text{spam}} = \frac{40}{20 + 40} = 0.67$$

$$P_{\text{ham}} = \frac{50}{50 + 20} = 0.71$$

take macro avg of PR pair → $\frac{0.67 + 0.71}{2} = 0.69$

if $0.69 \geq CL$:
approve

else :
improve.

Example 2:

Healthcare domain

$$s(\text{sick}) = 5 + 100 = 105$$

$$s(\text{healthy}) = 50 + 10 = 60$$

SL = 0.1
CL = 0.9

	healthy	sick
healthy	50 ₁	10 ₃
sick	5 ₂	100 ₄

→ area of focus

dataset is UNBALANCED

approach: Domain approach using PR pair

Mistakes:

① healthy predicted as sick ← (\$\$)

② sick predicted as healthy — NOT TOLERABLE

$$P_{\text{healthy}} = \frac{50}{50 + 5} = 0.909$$

$$R_{\text{sick}} = \frac{100}{5 + 100} = 0.95$$

$$\text{macroAvg} = \frac{0.909 + 0.95}{2} = 0.9295 \sim 0.93$$

is macroAvg $\geq$ CL $\longrightarrow$
0.93 $\geq$ 0.9 $\longrightarrow$ YES $\longrightarrow$ Approve.